FINS5545 | FINANCIAL MARKET DATA LITERACY

Systematic Fund Explorer

Systematic Multi-Asset Funds with News-Sentiment Analytics

Part B Report | Stations 3-4

Student ID: z5727282

Evidence period: 2021-2023 out-of-sample results

Prepared for academic assessment | August 2026

Executive summary

Systematic Fund Explorer puts the project's portfolio tests into one working app. It compares nine historical model funds: Equal Weight, Risk Parity and Minimum Variance across equity-only, cryptocurrency-only and combined universes. Every fund uses the same display of growth, return, volatility, Sharpe ratio, maximum drawdown and latest target weights. The underlying results all come from the same walk-forward out-of-sample process.

The clearest result is that the highest return was not automatically the best investor outcome. Combined Equal Weight earned 15.14% a year, with 21.60% volatility and a -27.87% maximum drawdown. Combined Risk Parity earned slightly less at 13.97%, but reduced volatility to 16.20% and drawdown to -19.47%, giving it the combined family's best Sharpe ratio of 0.888. Cryptocurrency returns were much higher, although drawdowns of about 75% to 82% would be difficult to tolerate, particularly for someone with a short horizon or limited loss capacity.

The sentiment experiment went the other way. A one-day-lagged VADER sector tilt lowered Equity Equal Weight's annualised return from 12.64% to 12.28% and its Sharpe ratio from 0.819 to 0.801. Turnover increased, and stronger tilts performed progressively worse. I kept this result because the evidence does not support presenting the signal as a successful product feature.

> **MAIN CONCLUSION** For this short historical sample, Combined Risk Parity gave the most convincing balance of return, volatility and drawdown. The app is useful as a comparison and learning tool, but the results are not forecasts and the prototype is not ready to operate as an investable service.

1. Product and investor value proposition

> **PRODUCT SCOPE** Nine completed systematic model funds across equity, cryptocurrency and combined universes, supported by walk-forward out-of-sample results and an equity headline-sentiment analysis.

1.1 Target user and problem

The app is designed for retail investors who understand basic financial terms but find it difficult to compare portfolio methods. Many investment websites focus on individual assets and recent price movements. This app starts with the fund. It shows how each model fund is built, how it performed during the test, how far it fell from a previous peak, and its latest target weights.

For these users, the important question is not only which fund earned the highest return, but how much volatility and drawdown came with it. The walk-forward method makes the comparison more credible: weights are estimated from earlier observations and then applied to later returns. The fact sheets connect this historical record to the latest model target weights. These are not live holdings, and the backtest is not a forecast. The app is therefore a comparison and learning tool, not investment advice.

1.2 Fund shelf

Asset family	Investable universe	Rolling window	Annualisation
Equity only	50 US large-cap equities	252 observations	252
Cryptocurrency only	10 cryptocurrencies	365 observations	365
Combined	50 equities + 10 cryptocurrencies	252 observations	252

Each family is implemented with Equal Weight, Risk Parity and long-only Minimum Variance, producing nine distinct historical model-fund prototypes.

1.3 Investor journey

- Compare like-for-like fund methods within an asset family.
- Open a fund fact sheet and inspect performance and current target holdings.
- Choose any two completed funds and test a hypothetical fixed allocation.
- Inspect sector sentiment, the approved fusion experiment, turnover and cost sensitivity.

The navigation follows the way I would investigate a fund. Users first compare methods using the same performance measures. They then open a fact sheet to inspect the growth path and target weights, before testing a hypothetical split between two funds. The sentiment and robustness pages come later because they support the main fund analysis rather than replace it.

Keeping these steps in one app removes the need to compare separate spreadsheets and charts, and makes the return-risk trade-offs easier to see. The scope of the prototype is intentionally narrow. It does not hold investor money, place trades, assess client suitability, charge fees or operate as a commercial fund.

2. Fund design and walk-forward methodology

2.1 Data, calendars and timing

I compute daily simple returns from adjusted-close prices in each asset's original panel. Cryptocurrency returns first follow the seven-day cryptocurrency calendar. For the combined funds, the completed cryptocurrency return series is aligned to observed equity trading dates, with no price or return forward-filling. Weekend observations remain in the cryptocurrency-only tests but do not add extra dates to the combined funds. Equity and combined funds therefore follow the equity trading calendar, while cryptocurrency-only funds retain the seven-day calendar.

Target weights are reset on the first observed trading day of each calendar month. Each target uses only the completed rolling window ending before the return to which that target applies. Under this timing convention, the first OOS date is 4 January 2021 for equity-only and combined funds, and 1 January 2021 for cryptocurrency-only funds.

2.2 Portfolio methods

Method	Objective	Implementation control
Equal Weight	1/N target	Transparent baseline; no covariance estimate
Risk Parity	Equal risk contribution	5% diagonal covariance shrinkage; deterministic solver checks
Minimum Variance	Minimise portfolio variance	SLSQP; long-only; weight-sum and objective validation

2.3 Evaluation assumptions

- Long-only, fully invested and no leverage.

- Zero annual risk-free rate for the reported Sharpe ratio.
- Saved fund returns are gross of transaction costs, fees and taxes.
- Separate 0 and 10 basis-point one-way transaction-cost checks use realised pre-trade turnover.
- Growth compounds daily returns geometrically; maximum drawdown is measured from the running wealth peak including the initial $1.

The lookback window matches each market's calendar: 252 complete observations for equity and combined funds, and 365 for cryptocurrency-only funds. Both windows represent approximately one year of evidence. This avoids creating weekend returns or discarding them before the cryptocurrency-only optimisation. The design can respond to changing conditions, but a one-year covariance estimate can still be noisy when the number of assets is large relative to the sample.

Monthly rebalancing balances responsiveness and turnover. Rebalancing more often keeps a portfolio closer to its estimated risk targets but usually increases turnover. Rebalancing less often reduces trading but allows risk contributions to drift. The portfolios are long-only, fully invested and use no leverage. These rules make the funds easier to interpret and avoid assumptions about short selling and borrowing, but they also prevent Minimum Variance from using short positions as hedges.

Equal Weight is a transparent benchmark with little reliance on estimated parameters. Risk Parity follows the equal-risk-contribution framework in Maillard, Roncalli and Teiletche (2010), balancing estimated risk contributions rather than allocating capital equally. Minimum Variance follows the Markowitz (1952) variance objective and is more sensitive to covariance-estimation errors. A 5% diagonal shrinkage is used to reduce instability, although it cannot remove estimation error. All Sharpe ratios use a zero annual risk-free rate for consistency. Gross results are checked against realised turnover and an illustrative one-way transaction cost of 10 basis points. This is a sensitivity test, not an estimate of every investor's actual trading cost.

3. Out-of-sample fund results

Appendix A, Table 1 reports the full performance comparison. In the equity family, Equal Weight produced the strongest growth and Sharpe ratio: 12.64% annualised return, 16.12% volatility, a 0.819 Sharpe ratio and a -20.25% maximum drawdown. Risk Parity reduced volatility to 14.53% and drawdown to -18.49%, but its return fell to 9.91% and its Sharpe ratio to 0.723. Minimum Variance recorded the lowest volatility at 12.67% and the shallowest drawdown at -15.28%, but returned only 5.50% with a 0.486 Sharpe ratio. In this sample, the extra reduction in variance cost too much return to improve risk-adjusted performance.

The cryptocurrency funds operated at a very different risk level. Risk Parity recorded the highest annual return in the family at 44.30% and the highest Sharpe ratio at 0.862. Equal Weight returned 40.50%, with volatility of 81.89%. Minimum Variance reduced volatility to 65.24% and maximum drawdown to -74.56%, but the loss remained severe. These results show why return alone is misleading: during the backtest, every cryptocurrency strategy fell roughly three-quarters to four-fifths below a previous peak.

The combined funds show the diversification trade-off most clearly. Equal Weight had the highest return at 15.14%, together with 21.60% volatility and a -27.87% maximum drawdown. Risk Parity retained most of that return at 13.97%, while reducing volatility to 16.20% and drawdown to -19.47%. This gave it the strongest Sharpe ratio in the combined family at 0.888. Minimum Variance returned 5.47%, with 12.70% volatility and a 0.483 Sharpe ratio. Its lower volatility did not make up for the return that was lost.

These figures describe one sample and should not be treated as a fixed or generalisable ranking. Equal Weight is simple but accepts more variation. Risk Parity produced the strongest risk-adjusted result in the combined family. Minimum Variance reduced measured risk but returned much less and remained sensitive to covariance-estimation noise and concentration. The results do not identify a fund that is suitable for every investor.

3.1 Drawdown and risk-return evidence

Appendix A, Figure 2 shows that the losses were not isolated one-day shocks. Equity and combined funds experienced their longest stress during the broad market decline in 2022, when Combined Equal Weight reached a -27.87% maximum drawdown. Combined Risk Parity had a shallower trough. This pattern is consistent with spreading estimated risk contributions more evenly, although the backtest does not establish that mechanism as the cause. Minimum Variance had still shallower drawdowns, but at the cost of much weaker growth.

Cryptocurrency portfolios delivered a far more difficult holding experience for investors. Maximum drawdowns spanned -74.56% to -81.57%. Impressive full-period annualised returns therefore co-existed with extended stretches where portfolio value stayed below prior wealth peaks. Both how deep losses fall and how long they persist carry practical significance. A strong final portfolio value provides little reassurance when the path to that outcome clashes with an investor's liquidity requirements, risk tolerance or investment time horizon. While the visual outputs pinpoint the timing and magnitude of historical losses, they cannot pin down the underlying economic drivers.

Appendix A, Figure 3 separates the asset families because one set of axes would compress the equity and combined funds beside the much more volatile cryptocurrency funds. The faceted view makes differences within each family easier to read and avoids implying that all annualised returns are directly comparable. Equity and combined results use 252 observed trading days for annualisation, while cryptocurrency-only results use 365 calendar days. Each rate matches its own return series, but the calendar and market-structure differences still matter when the panels are compared.

In the combined panel, Risk Parity lies to the left of Equal Weight with only a modest reduction in return, which explains its higher Sharpe ratio. Minimum Variance lies further left but much lower, showing that minimising estimated variance is not the same as maximising realised risk-adjusted return. The cryptocurrency funds combine high returns with extreme volatility and drawdown. The figure is therefore best used to compare trade-offs within each asset family, not to create one universal ranking across all nine funds.

3.2 Portfolio construction and implementation sensitivity

Complete monthly target weight histories for all model funds are saved within results/data/fund_weights.csv. Realised one-way turnover metrics and the illustrative 10-basis-point transaction-cost sensitivity test appear in two separate files: results/tables/fund_turnover.csv and fund_transaction_cost_check.csv.

Combined Risk Parity does not represent a fixed 1/60 allocation. Monthly target weights shifted over time alongside updates to rolling volatility and correlation estimates, as visualised in Appendix A, Figure 4. For chart clarity, the largest individual weights are plotted explicitly, while smaller positions are aggregated under "other holdings". This strategy distributes estimated risk across assets instead of splitting capital evenly. Consequently, lower-volatility securities may attract larger capital allocations, preventing highly volatile cryptocurrency positions from overwhelming overall portfolio-risk estimates. Even so, examining the full holdings dataset remains necessary to evaluate concentration risk at any given rebalancing point.

Turnover metrics highlight material practical distinctions across portfolio construction approaches. Combined Risk Parity recorded total one-way turnover of roughly 1.32, against 1.35 for Combined Equal Weight. By comparison, Combined Minimum Variance reached 5.08. Applying the hypothetical 10-basis-point one-way transaction cost produces only moderate downward adjustments to net annualised returns for most funds. Minimum Variance sees the greatest performance impact here, given its optimal weights tend to shift more aggressively between rebalancing cycles. Turnover is computed relative to drifted pre-trade weights, with the initial portfolio setup step excluded from calculations. Charts and fact-sheet outputs display monthly target allocations only. Real-world holdings would naturally drift in the intervals between rebalances, and diverge further once practical execution frictions are taken into account.

4. Sector sentiment and structured-unstructured fusion

4.1 Standalone sector sentiment index

VADER compound scores are calculated only from the supplied equity headlines, with casing, punctuation and negation left intact. I first average headlines at the ticker-day level and then take an equal average across observed tickers in each sector. The validation grid has 10 sectors and 10,060 possible sector-days. Headlines are present for 9,832 of them, leaving 228 missing observations. Coverage is not even: Real Estate and Materials each contribute 66 missing days, and Utilities contributes 44. If a sector has no observed headline on a day, I leave the value missing rather than treating it as neutral.

Figure 5 in Appendix A shows the resulting index. It captures the tone of the supplied equity headlines, not a company's financial health or intrinsic value. VADER assigns each headline a compound score between -1 and +1, which I then aggregate by ticker-day and sector. Retaining casing, punctuation and negation follows the rule-based design in Hutto and Gilbert (2014). A zero has a specific meaning: an observed headline was neutral or contained no recognised valence. A missing value means that no eligible headline was observed. Treating those cases as the same would insert information that the dataset does not contain.

There are several reasons to treat this index cautiously. Headlines are short, can be ambiguous, and are selected by publishers rather than sampled evenly. VADER was built as a general rule-based model for social-media text, so finance-specific expressions may be missed. Averaging can hide disagreement between firms, and uneven coverage changes both the number and identity of tickers behind a sector score. I therefore use the index as a descriptive measure of the supplied news, not as a causal return predictor. It is also restricted to equities: the headlines map to the 50 equity tickers and their sectors, and applying them directly to cryptocurrency would require a link that is not supported by the data.

4.2 Look-ahead-safe fusion experiment

Appendix A, Table 2 and Figure 6 compare the base strategy with the fusion test. The approved version uses the previous observed equity day's sector score and applies a 0.5 tilt to Equal Weight targets. The lag matters: a headline assigned to day t cannot influence the portfolio until day t+1. If the earlier sector score is missing, the multiplier remains 1.0; the process does not create a neutral score or an active tilt. On the results, fusion did not improve the base fund. Annualised return fell from 12.64% to 12.28%, the Sharpe ratio from 0.819 to 0.801, and terminal growth from $1.427 to $1.414. Volatility improved by just 0.04 percentage points, while maximum drawdown became slightly worse.

The robustness checks point in the same direction. Increasing the tilt from 0.25 to 1.00 steadily reduced annualised return, from 12.46% to 11.95%, and lowered the Sharpe ratio from 0.810 to 0.784. Total one-way turnover rose from 0.953 for the base strategy to 1.379 at the approved 0.5 tilt and 2.079 at a 1.0 tilt. After a 10 bps cost, the approved fusion returned 12.23% compared with 12.61% for the base. Costs widened the gap rather than rescuing the signal.

A failed signal is still a result. In this sample, the simple lagged headline tilt lowered return and Sharpe ratio while increasing turnover. That does not mean every form of sentiment information is useless; it means that turning up this particular tilt is not a credible next step. A better follow-up would change the language model, exercise tighter control over news sources and test a longer aggregation window.

5. Innovation and deployed investor experience

5.1 Implemented extensions beyond the minimum

- Nine-fund shelf rather than only two combined methods.
- Long-only Minimum Variance alongside Equal Weight and Risk Parity.
- Native-calendar cryptocurrency backtests with 365-observation estimation and annualisation.
- Pre-trade turnover measurement and fund-level transaction-cost sensitivity.
- Sentiment-fusion turnover, cost and tilt-strength robustness checks.
- Dynamic nine-fund fact sheets and an any-two-fund historical allocation lab.
- Asset-family faceting to avoid visually conflating equity and seven-day cryptocurrency calendars.

The largest extension was moving from a two-method combined-fund baseline to a shelf of nine funds. I applied three portfolio methods to equity-only, cryptocurrency-only and combined universes, while retaining the appropriate calendar for each family and reporting the same decision measures throughout. This broader comparison materially changes what the app can show. Combined Risk Parity produced the strongest Sharpe ratio in the combined family (0.888), whereas Minimum Variance reduced the reported risk measures but delivered weak realised Sharpe ratios. The result depends not just on the method, but also on the asset family and the kind of risk an investor is trying to control.

The second extension asks whether the more complicated methods still look attractive once implementation is considered. Using the same OOS evidence, I measured pre-trade turnover, an illustrative one-way cost of 10 bps and five strengths of the sentiment tilt. Combined Minimum Variance recorded total one-way turnover of about 5.08, compared with 1.35 for Combined Equal Weight and 1.32 for Combined Risk Parity. Sentiment performance also deteriorated as the tilt increased. These checks matter because a lower estimated variance or a more elaborate signal can still produce a worse net outcome.

Dynamic fact sheets, the any-two-fund allocation lab and separate method notes make the evidence easier to explore. The allocation lab applies one historical fixed split and does not rebalance the two selected funds after the initial allocation. It is intended to illustrate scenarios, not to optimise a portfolio or make a recommendation.

5.2 Deployment and operating boundary

The deployed Streamlit app reads committed, precomputed files from results/. It does not download raw data, refit portfolios, run VADER or rebuild exhibits during a user's session. This design keeps the public app responsive on Streamlit Community Cloud and ensures that its figures and tables are drawn from the same evidence submitted with the report.

Live prototype: https://z5727282-projectb.streamlit.app/

I checked the public version from the opening comparison page through the fact sheets, allocation lab, fund analytics, sector sentiment, fusion robustness and method notes. The displayed figures matched the committed report tables. The public repository is https://github.com/EdwarrdsLi/z5727282_projectB. Because Streamlit Community Cloud deploys from the repository entry point, committing the result artefacts also makes the displayed evidence reproducible (Streamlit, 2026).

Analysis and deployment are separated on purpose. Portfolio estimation and VADER processing happen before deployment; the public app only loads the saved outputs. This makes the app faster and prevents the underlying evidence from changing during a session. As a final check, I opened both the GitHub repository and the Streamlit app while logged out and confirmed that they were publicly accessible.

6. Critical reflection and recommendations

6.1 What worked, what did not, and why

The completed prototype contains nine walk-forward funds, their fact sheets, a sector sentiment index, a lagged fusion test and a public app. For me, the most useful finding was not the highest return. It was seeing how much the investor's path changed under each construction method. Combined Risk Parity kept most of the return from Combined Equal Weight while reducing volatility and drawdown, which left it with the best Sharpe ratio in the combined family.

There are several reasons not to overstate that conclusion. The OOS period covers only 2021-2023 and therefore contains a limited set of market conditions. A fixed universe of 50 large-cap equities and 10 cryptocurrencies is not the full investable market and may include selection or survivorship effects. I have not reported confidence intervals, bootstrap results or formal significance tests, so small differences between methods could simply be sample noise. Cryptocurrency drawdowns remained extreme under every method. Minimum Variance relied on an estimated covariance matrix and generated much higher turnover, making its defensive appearance less stable in practice. Plain VADER was never designed as a finance-specific return model, and the fusion result worsened with stronger tilts. Finally, the headline figures are gross of management fees, bid-ask spreads, market impact, taxes and liquidity constraints.

A real fund would need far more than a longer backtest. Among other things, it would require validated execution rules, instrument-eligibility checks, stale-price controls, liquidity and capacity limits, custody, compliance, ongoing monitoring and a process for investor suitability. The present evidence only compares historical designs; it says nothing certain about future performance. Even so, the unsuccessful sentiment test has value because it stops an appealing story from being promoted when the numbers do not support it.

6.2 Three concrete recommendations

Recommendation	Implementation and evaluation
1. Stabilise portfolio construction	Apply stronger covariance shrinkage and position caps, then compare rolling-window alternatives in a nested walk-forward test. Evaluate OOS Sharpe, drawdown, concentration and turnover rather than return alone.
2. Build finance-specific sentiment	Extend VADER with a documented finance lexicon, source-quality weights and a longer aggregation horizon. Freeze choices before the next OOS period, retain the one-day lag, and compare information coefficients, ranking stability, turnover and net performance with both baselines.
3. Add production controls	Model spreads, nonlinear market impact, management fees and liquidity limits; add stale-data, drift and drawdown monitoring. Require logged-out deployment and clean-checkout reproducibility tests, and progress only if the strategy remains feasible after costs.

6.3 Final conclusion

Systematic Fund Explorer achieves its narrower goal: a transparent, reproducible prototype for comparing historical model funds. In the combined universe, Risk Parity produced the strongest balance of return, volatility and drawdown. Cryptocurrency funds earned much more, but their peak-to-trough losses would be difficult for investors with limited loss tolerance or a shorter horizon. Minimum Variance lowered volatility while giving up substantial return. The sentiment experiment produced a less marketable but still useful finding: the simple lagged VADER tilt did not improve performance.

These findings are still uncertain. The sample is short, the cost assumption is illustrative, covariance estimates can move sharply, and a headline signal may behave differently in another regime. My next steps would be to extend the OOS period, strengthen covariance and concentration controls, build and validate a finance-specific sentiment measure, and model execution and monitoring more realistically. Until then, the app should continue to label its results as historical analysis rather than personalised investment advice.

7. AI workflow and quality control

I did not ask AI to decide what the project should contain. Before implementation began, I provided a detailed workflow covering the fund families, walk-forward OOS design, calendar rules, sentiment method, fusion controls, innovation requirements, investor journey and deployment boundaries. I also specified the main risks that had to be checked, including look-ahead bias, incorrect annualisation, Risk Parity collapsing towards Equal Weight, unlagged sentiment, hidden negative results and cloud recomputation. AI mainly helped turn these instructions into an initial code structure, test framework and report outline.

I made the final decisions about the project design. I approved the final nine-fund shelf, the use of Equal Weight, Risk Parity and Minimum Variance across three asset families, the native cryptocurrency calendar and the separation of gross performance from transaction-cost checks. I also decided how the app should present the evidence and which additional robustness checks were needed. When the sentiment tilt reduced return and Sharpe ratio, I kept the result instead of changing the specification until it looked successful.

After each AI-assisted stage, I checked the output against the project requirements and saved evidence. I reviewed the date logic, target-weight constraints, 252/365 annualisation, sentiment lag, missing-news treatment, turnover calculations and transaction-cost arithmetic. I compared the reported figures with the saved CSV files and used focused tests to check the implementation. The final interpretation was based on those verified results rather than on the AI's description of them.

AI output also required correction. For example, an early fix to a horizontal chart still placed the title on the wrong physical axis. I found the problem through a screenshot, corrected the mapping and reran the relevant tests. I also identified truncated metric labels and a Streamlit Cloud theme problem. These issues were fixed without changing the calculations.

For the written report, AI provided an initial structure and drafting framework. I then reviewed the report section by section, checked its claims against the project artifacts, removed or changed wording that did not reflect my understanding, and substantially rewrote the analysis, limitations, recommendations and final interpretation. AI supported the process, but the project choices, verification decisions, final arguments and submitted wording remain my responsibility.

References

Hutto, C.J. and Gilbert, E. (2014) 'VADER: A parsimonious rule-based model for sentiment analysis of social media text', Proceedings of the International AAAI Conference on Web and Social Media, 8(1), pp. 216-225. https://doi.org/10.1609/icwsm.v8i1.14550

Maillard, S., Roncalli, T. and Teiletche, J. (2010) 'The properties of equally weighted risk contribution portfolios', Journal of Portfolio Management, 36(4), pp. 60-70. https://doi.org/10.3905/jpm.2010.36.4.060

Markowitz, H. (1952) 'Portfolio selection', Journal of Finance, 7(1), pp. 77-91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

Streamlit (2026) 'Deploy your app on Community Cloud'. Available at: https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/deploy (Accessed: 13 August 2026).

UNSW Business School (2026) FINS5545 Financial Market Data Literacy: FinTech Project 2026 - Project Brief. Supplied course document in the project repository.

Project artefacts and supplied data

UNSW Business School (2026) FINS5545 course data bundle: adjusted-close price panels, security master and supplied equity headlines, accessed through src/data_access.py.

Computed project evidence: results/data/fund_returns.csv, results/data/fund_weights.csv, results/data/sector_sentiment_index.csv, and the verified tables and figures under results/.

Appendix A. Required exhibits

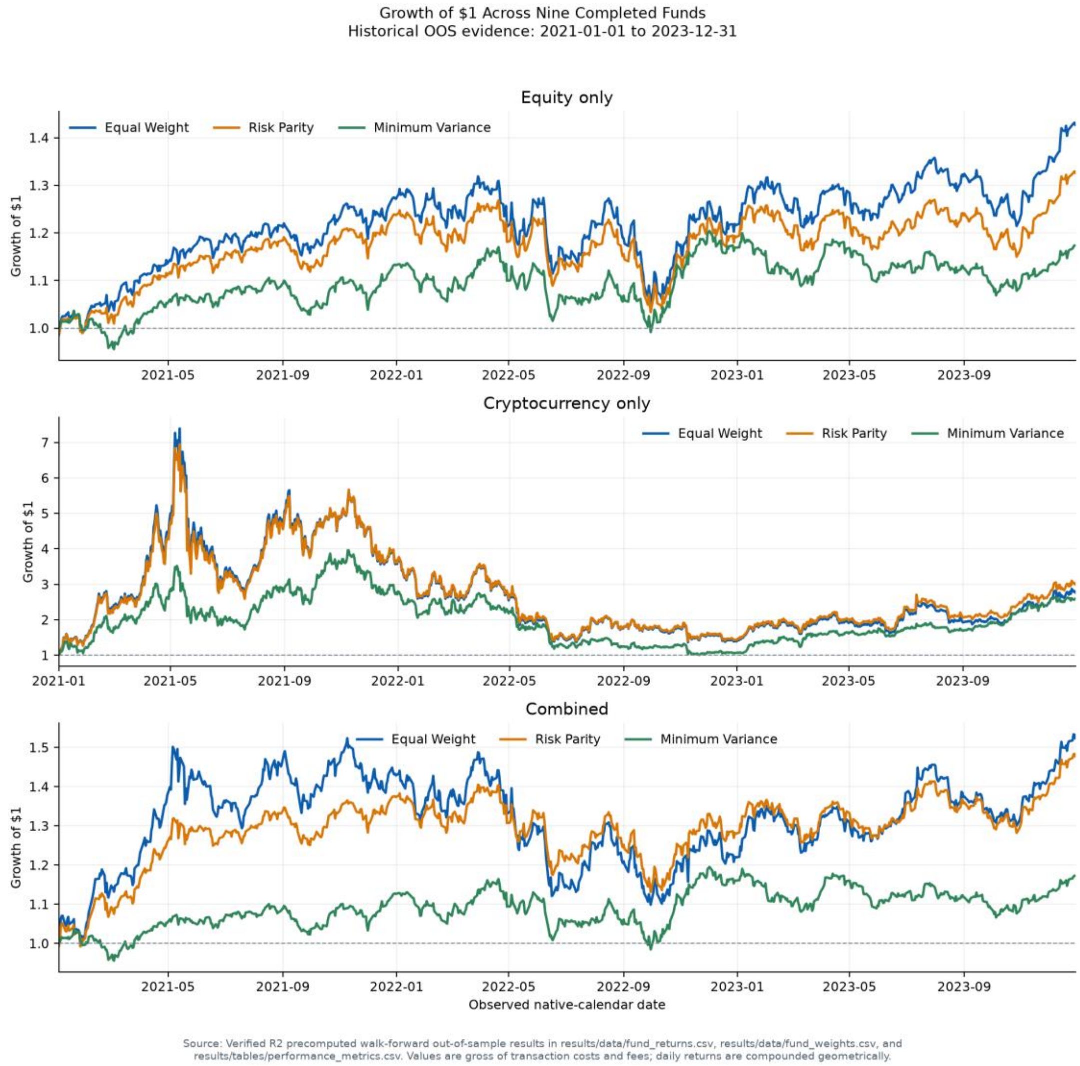
This appendix contains the required Part B exhibits. Sections 3 and 4 interpret each table and figure. All values are historical, precomputed walk-forward out-of-sample evidence.

Table 1. Verified OOS performance across nine completed funds

Fund	Ann. return	Ann. volatility	Sharpe	Max drawdown	Growth of $1
Equity EW	12.64%	16.12%	0.819	-20.25%	$1.427
Equity RP	9.91%	14.53%	0.723	-18.49%	$1.326
Equity MinVar	5.50%	12.67%	0.486	-15.28%	$1.174
Crypto EW	40.50%	81.89%	0.829	-81.57%	$2.774
Crypto RP	44.30%	79.90%	0.862	-79.88%	$3.005
Crypto MinVar	37.39%	65.24%	0.814	-74.56%	$2.593
Combined EW	15.14%	21.60%	0.761	-27.87%	$1.524
Combined RP	13.97%	16.20%	0.888	-19.47%	$1.478
Combined MinVar	5.47%	12.70%	0.483	-15.42%	$1.173

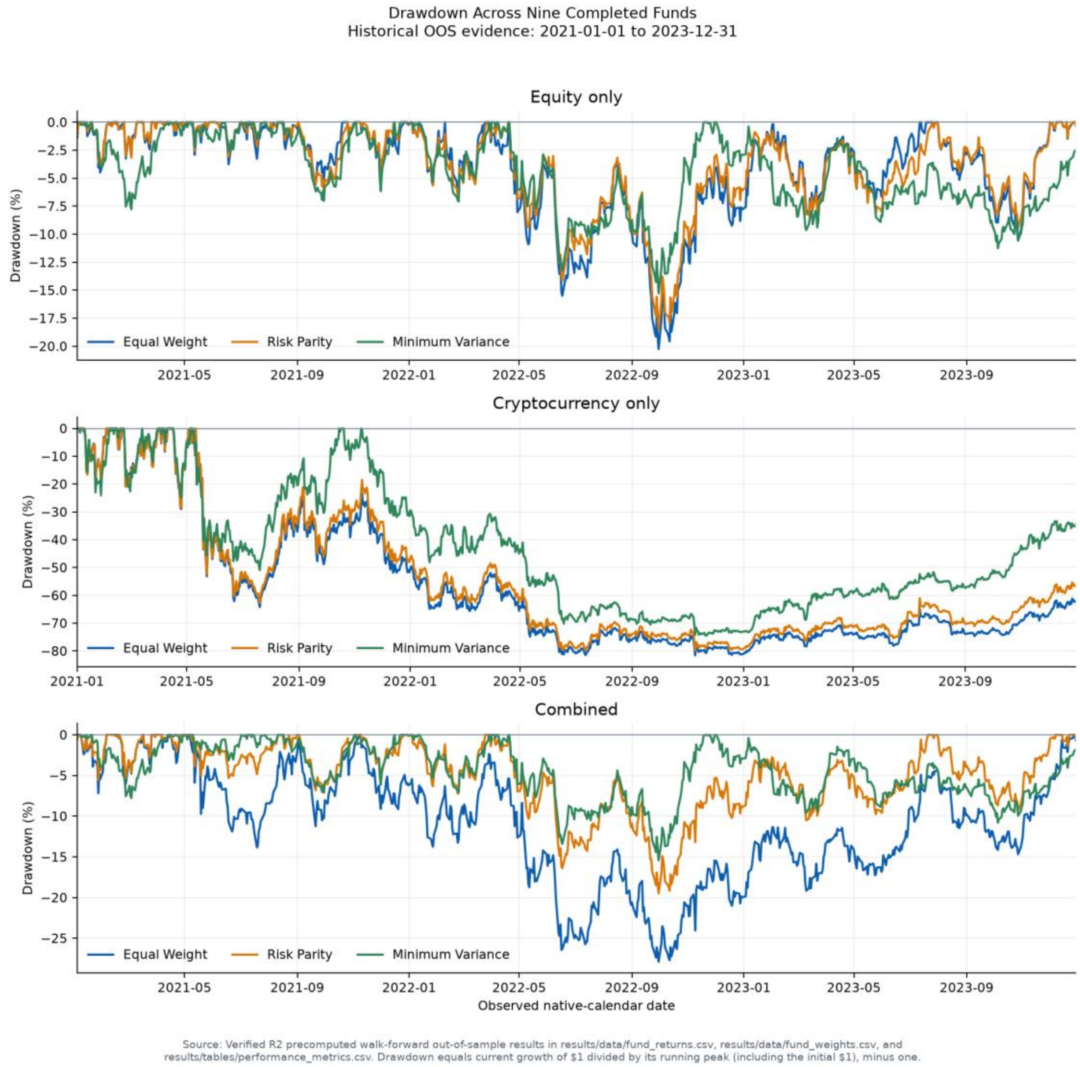
Source: results/tables/performance_metrics.csv; zero annual risk-free rate; gross returns. EW = Equal Weight; RP = Risk Parity; MinVar = Minimum Variance.

Figure 1. Growth of $1 across the nine completed funds, faceted by asset family



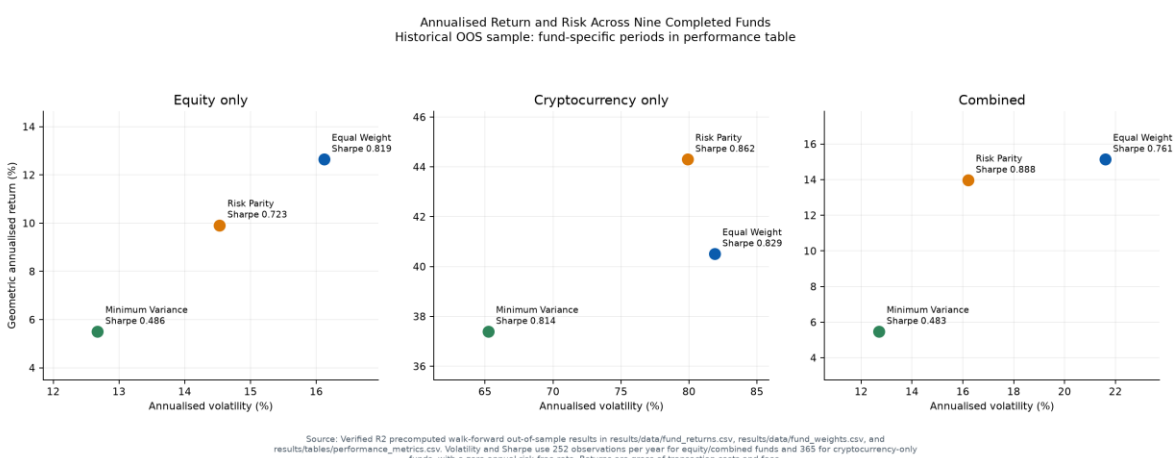
Source: Verified precomputed OOS returns in results/data/fund_returns.csv. The displayed span begins with the cryptocurrency funds on 2021-01-01; equity and combined funds begin on 2021-01-04.

Figure 2. Historical drawdowns across the nine completed funds



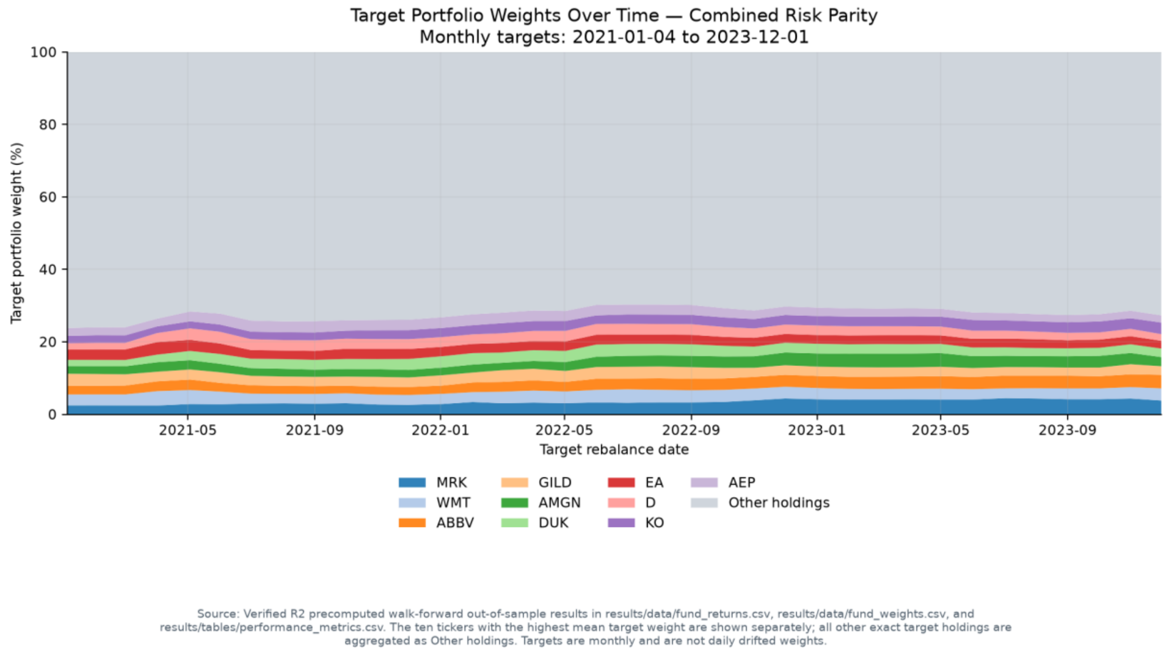
Source: Verified precomputed OOS growth paths; drawdown includes the initial $1 peak.

Figure 3. Annualised return and risk across asset families and methods



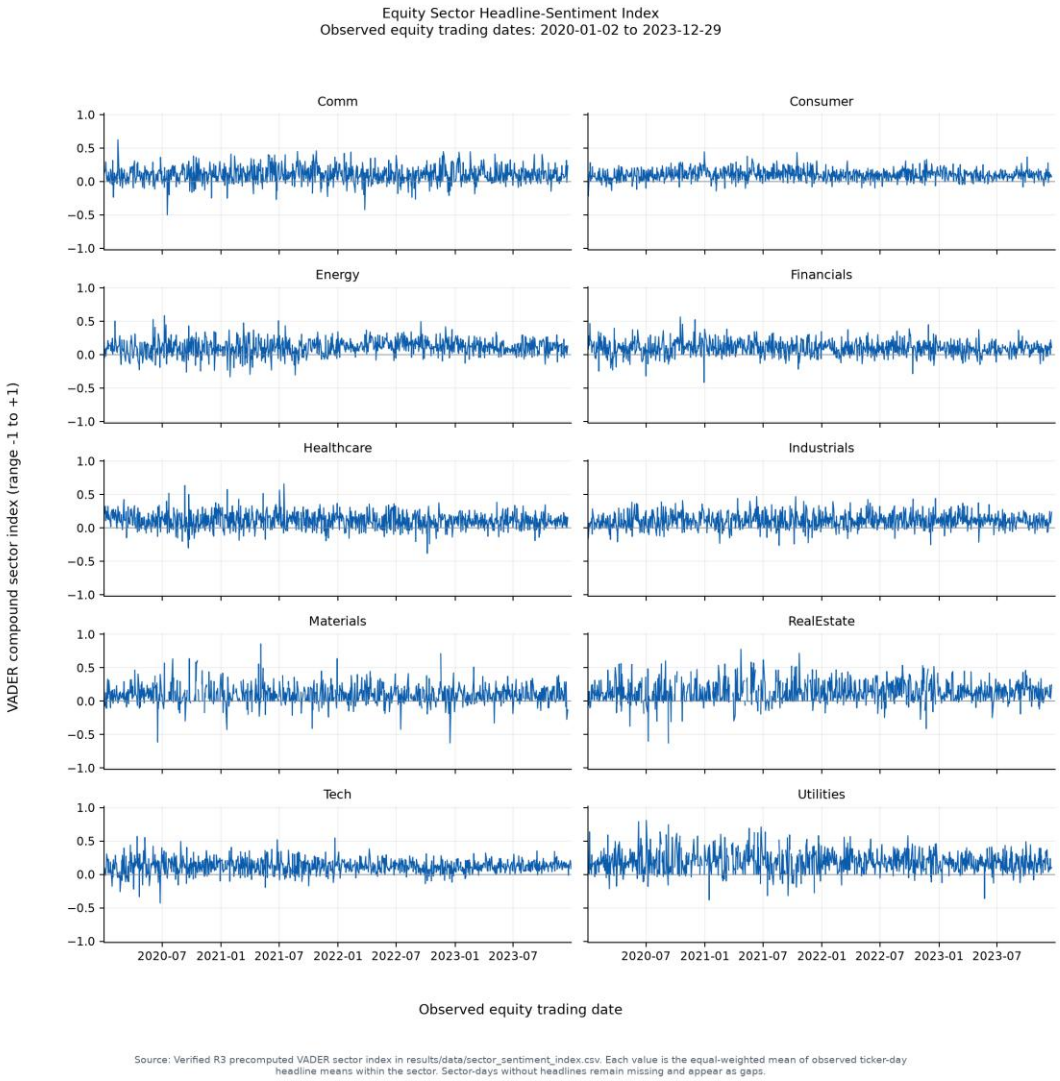
Source: results/tables/performance_metrics.csv; 252/365 annualisation matched to fund calendar.

Figure 4. Combined Risk Parity monthly target weights over time



Source: results/data/fund_weights.csv; ten highest mean target weights shown separately.

Figure 5. Equity sector headline-sentiment index



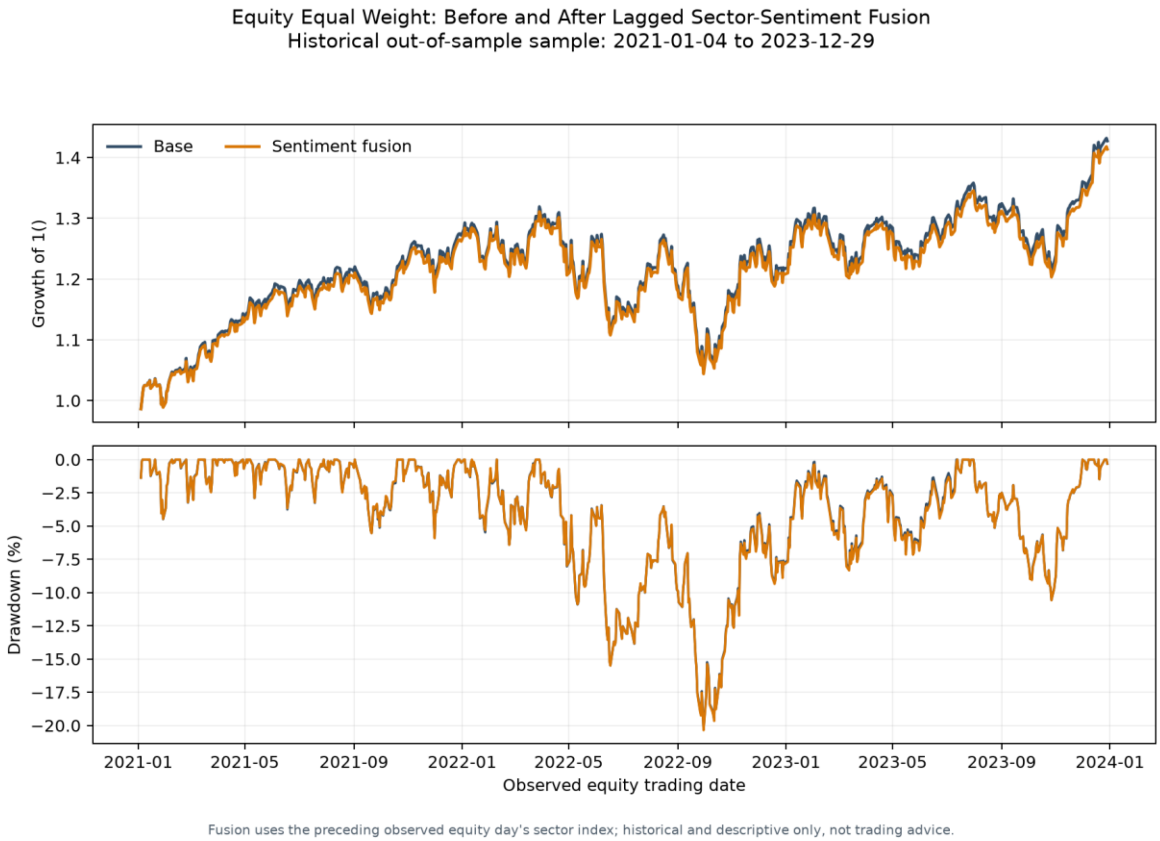
Source: Verified VADER index in results/data/sector_sentiment_index.csv; 2020-01-02 to 2023-12-29. Coverage: 10 sectors, 9,832 observed sector-days and 228 no-headline sector-days.

Table 2. Equity Equal Weight before and after the approved sentiment tilt

Strategy	Ann. return	Ann. volatility	Sharpe	Max drawdown	Growth
Equity Equal Weight	12.64%	16.12%	0.819	-20.25%	$1.427
Equity Equal Weight + Sector Sentiment	12.28%	16.08%	0.801	-20.36%	$1.414

Source: results/tables/sentiment_fusion_comparison.csv; identical OOS sample.

Figure 6. Growth of $1 before and after sentiment fusion

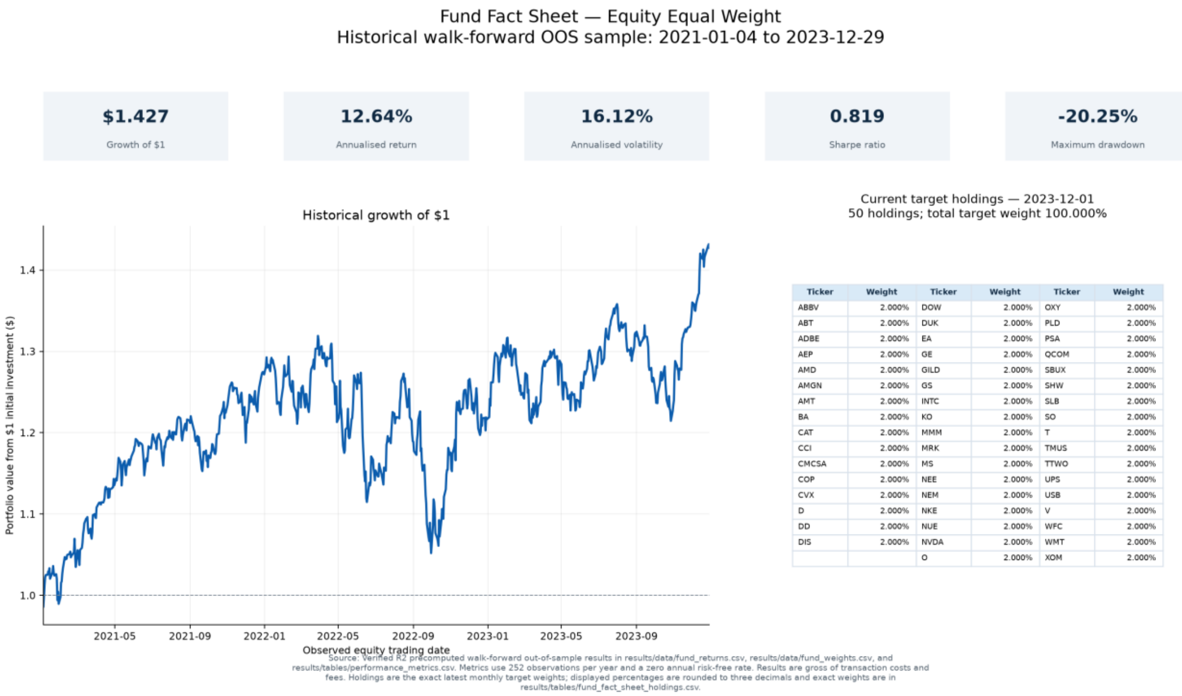


Source: Verified precomputed comparison; one observed equity-day signal lag and 0.5 approved tilt.

Appendix B. Fund fact sheets

The following fact sheets report the exact verified OOS metrics and latest non-zero monthly target holdings for each completed asset-family/method fund. These are target weights, not live holdings or recommendations.

Appendix B1. Equity Equal Weight



Ticker	Weight	Ticker	Weight	Ticker	Weight
ABBV	2.000%	DOW	2.000%	OXY	2.000%
ABT	2.000%	DUK	2.000%	PLD	2.000%
ADBE	2.000%	EA	2.000%	PSA	2.000%
AEP	2.000%	GE	2.000%	QCOM	2.000%
AMD	2.000%	GILD	2.000%	SBUX	2.000%
AMGN	2.000%	GS	2.000%	SHW	2.000%
AMT	2.000%	INTC	2.000%	SLB	2.000%
BA	2.000%	KO	2.000%	SO	2.000%
CAT	2.000%	MMM	2.000%	T	2.000%
CCI	2.000%	MRK	2.000%	TMUS	2.000%
CMCSA	2.000%	MS	2.000%	TTWO	2.000%
COP	2.000%	NEE	2.000%	UPS	2.000%
CVX	2.000%	NEM	2.000%	USB	2.000%
D	2.000%	NKE	2.000%	V	2.000%
DD	2.000%	NUE	2.000%	WFC	2.000%
DIS	2.000%	NVDA	2.000%	WMT	2.000%
		O	2.000%	XOM	2.000%

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B2. Equity Risk Parity

Fund Fact Sheet — Equity Risk Parity
Historical walk-forward OOS sample: 2021-01-04 to 2023-12-29

$1.326	9.91%	14.53%	0.723	-18.49%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
50 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
MRK	4.141%	NEM	2.027%	WFC	1.582%
ABBV	3.880%	XOM	2.012%	AMT	1.577%
WMT	3.810%	D	1.951%	BA	1.573%
TMUS	3.449%	PSA	1.946%	CAT	1.570%
KO	3.448%	GE	1.939%	SLB	1.560%
AMGN	2.712%	CMCSA	1.856%	CCI	1.528%
GILD	2.594%	OXY	1.812%	NKE	1.526%
ABT	2.502%	NEE	1.797%	MS	1.520%
DUK	2.471%	SBUX	1.766%	INTC	1.469%
T	2.470%	TTWO	1.745%	MMM	1.463%
V	2.399%	COP	1.699%	AMD	1.438%
EA	2.371%	DOW	1.681%	ADBE	1.435%
SO	2.335%	SHW	1.664%	NVDA	1.395%
O	2.223%	GS	1.652%	PLD	1.356%
AEP	2.094%	UPS	1.648%	NUE	1.308%
CVX	2.062%	DD	1.634%	QCOM	1.284%
		DIS	1.597%	USB	1.029%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 252 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B3. Equity Minimum Variance

Fund Fact Sheet — Equity Minimum Variance
Historical walk-forward OOS sample: 2021-01-04 to 2023-12-29

$1.174	5.50%	12.67%	0.486	-15.28%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
21 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
KO	18.271%	EA	4.320%	NVDA	1.151%
WMT	17.354%	T	3.490%	NEM	1.089%
MRK	10.107%	CVX	2.723%	AMD	0.929%
ABBV	8.540%	OXY	2.589%	DD	0.923%
O	7.957%	ABT	2.291%	DUK	0.798%
V	7.915%	AMGN	1.410%	INTC	0.274%
TMUS	6.504%	GE	1.305%	DOW	0.060%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 252 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B4. Cryptocurrency Equal Weight

Fund Fact Sheet — Cryptocurrency Equal Weight
Historical walk-forward OOS sample: 2021-01-01 to 2023-12-31

$2.774	40.50%	81.89%	0.829	-81.57%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
10 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
ADA-USD	10.000%	EOS-USD	10.000%	LTC-USD	10.000%
BCH-USD	10.000%	ETC-USD	10.000%	TRX-USD	10.000%
BTC-USD	10.000%	ETH-USD	10.000%	XLM-USD	10.000%
				XRP-USD	10.000%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 365 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B5. Cryptocurrency Risk Parity

Fund Fact Sheet — Cryptocurrency Risk Parity
Historical walk-forward OOS sample: 2021-01-01 to 2023-12-31

$3.005	44.30%	79.90%	0.862	-79.88%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
10 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
TRX-USD	15.489%	EOS-USD	9.106%	ADA-USD	8.840%
BTC-USD	13.291%	BCH-USD	9.010%	ETC-USD	8.695%
ETH-USD	11.686%	LTC-USD	8.944%	XLM-USD	7.664%
				XRP-USD	7.274%

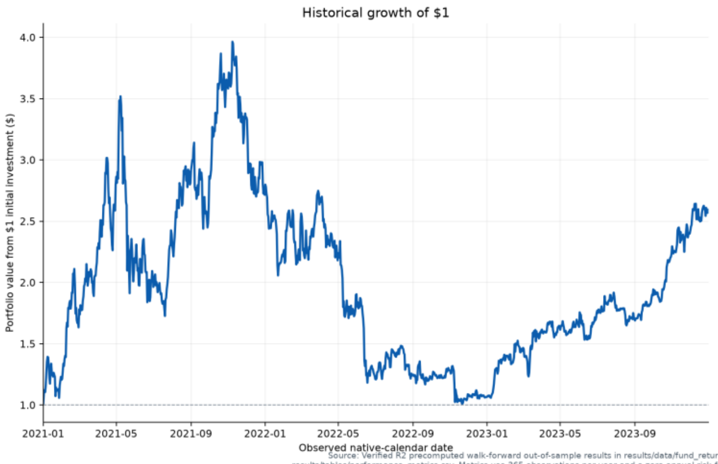
Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 365 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B6. Cryptocurrency Minimum Variance

Fund Fact Sheet — Cryptocurrency Minimum Variance
Historical walk-forward OOS sample: 2021-01-01 to 2023-12-31

$2.593	37.39%	65.24%	0.814	-74.56%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown



Current target holdings — 2023-12-01
3 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
TRX-USD	48.744%	BTC-USD	43.341%	ETH-USD	7.914%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 365 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B7. Combined Equal Weight

Fund Fact Sheet — Combined Equal Weight
Historical walk-forward OOS sample: 2021-01-04 to 2023-12-29

$1.524	15.14%	21.60%	0.761	-27.87%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
60 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
ABBV	1.667%	DUK	1.667%	OXY	1.667%
ABT	1.667%	EA	1.667%	PLD	1.667%
ADA-USD	1.667%	EOS-USD	1.667%	PSA	1.667%
ADBE	1.667%	ETC-USD	1.667%	QCOM	1.667%
AEP	1.667%	ETH-USD	1.667%	SBUX	1.667%
AMD	1.667%	GE	1.667%	SHW	1.667%
AMGN	1.667%	GILD	1.667%	SLB	1.667%
AMT	1.667%	GS	1.667%	SO	1.667%
BA	1.667%	INTC	1.667%	T	1.667%
BCH-USD	1.667%	KO	1.667%	TMUS	1.667%
BTC-USD	1.667%	LTC-USD	1.667%	TRX-USD	1.667%
CAT	1.667%	MMM	1.667%	TTWO	1.667%
CCI	1.667%	MRK	1.667%	UPS	1.667%
CMCSA	1.667%	MS	1.667%	USB	1.667%
COP	1.667%	NEE	1.667%	V	1.667%
CVX	1.667%	NEM	1.667%	WFC	1.667%
D	1.667%	NKE	1.667%	WMT	1.667%
DD	1.667%	NUE	1.667%	XLM-USD	1.667%
DIS	1.667%	NVDA	1.667%	XOM	1.667%
DOW	1.667%	O	1.667%	XRP-USD	1.667%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 252 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B8. Combined Risk Parity

Fund Fact Sheet — Combined Risk Parity
Historical walk-forward OOS sample: 2021-01-04 to 2023-12-29

$1.478	13.97%	16.20%	0.888	-19.47%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
60 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
MRK	3.790%	GE	1.736%	NKE	1.356%
ABBV	3.725%	NEE	1.716%	INTC	1.312%
WMT	3.367%	CMCSA	1.679%	TRX-USD	1.302%
KO	3.196%	SBUX	1.633%	MMM	1.295%
TMUS	3.161%	OXY	1.621%	ADBE	1.260%
AMGN	2.519%	COP	1.549%	BTC-USD	1.246%
GILD	2.377%	DOW	1.529%	PLD	1.238%
DUK	2.354%	UPS	1.520%	AMD	1.237%
ABT	2.289%	GS	1.516%	NVDA	1.185%
T	2.214%	TTWO	1.495%	NUE	1.152%
SO	2.169%	SHW	1.481%	QCOM	1.112%
EA	2.145%	DD	1.453%	ETH-USD	1.094%
O	2.106%	CCI	1.437%	USB	0.917%
V	2.099%	WFC	1.420%	BCH-USD	0.891%
AEP	1.985%	AMT	1.419%	LTC-USD	0.853%
CVX	1.916%	BA	1.408%	EOS-USD	0.821%
XOM	1.878%	SLB	1.408%	ADA-USD	0.792%
D	1.829%	CAT	1.404%	ETC-USD	0.712%
PSA	1.798%	DIS	1.373%	XLM-USD	0.693%
NEM	1.774%	MS	1.362%	XRP-USD	0.681%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 252 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.

Appendix B9. Combined Minimum Variance

Fund Fact Sheet — Combined Minimum Variance
Historical walk-forward OOS sample: 2021-01-04 to 2023-12-29

$1.173	5.47%	12.70%	0.483	-15.42%
Growth of $1	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown

Current target holdings — 2023-12-01
22 holdings; total target weight 100.000%

Ticker	Weight	Ticker	Weight	Ticker	Weight
KO	17.954%	EA	4.134%	TRX-USD	1.263%
WMT	16.609%	T	3.348%	BTC-USD	1.235%
MRK	9.980%	CVX	2.969%	NVDA	1.079%
ABBV	8.677%	OXY	2.392%	DUK	1.013%
O	7.966%	ABT	2.083%	NEM	0.879%
V	7.466%	GE	1.445%	AMD	0.790%
TMUS	6.589%	AMGN	1.304%	DD	0.559%
				INTC	0.265%

Source: Verified R2 precomputed walk-forward out-of-sample results in results/data/fund_returns.csv, results/data/fund_weights.csv, and results/tables/performance_metrics.csv. Metrics use 252 observations per year and a zero annual risk-free rate. Results are gross of transaction costs and fees. Holdings are the exact latest monthly target weights; displayed percentages are rounded to three decimals and exact weights are in results/tables/fund_fact_sheet_holdings.csv.

Source: Verified precomputed fund returns, metrics and latest target weights under results/.